



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

RELAY TEST SET · THREE PHASE

RTS-33

Three-phase, two-unit relay test set - 0-360° phase angle, 0-50 A current, 415V 3-phase input. For distance, directional, and differential relay testing.

0-360° PHASE

Three-Phase Relay Test Set

A three-phase relay test set for testing distance, over-current, over/under-voltage, earth fault, and directional-element relays - a two-unit system (current/voltage source + a dedicated phase-shifter unit) giving full 0-360° phase angle control between voltage and current, for genuine directional and distance relay testing not possible with single-phase sets.

- **0-360° phase angle** - coarse 60° steps + fine ±30°
- **0-50A across 5 taps** - 0-100% regulation each
- **Phase-to-phase fault mode** - selected phase currents in opposition
- **Digital timer** - 0-9.999s or 0-99.99s, two ranges
- **Phase sequence check** - built-in, before testing

Manufactured in India since 1963 - trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.



- P 0-360°** Phase angle, coarse + fine control
- A 0-50A** 5 continuously variable taps
- V 415V, 3-Phase** Input, 250 VA burden

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Input	415 V ±10%, 3 phase, 4 wire, 50 Hz A.C.
Burden	250 VA (current and voltage circuits)
Test Current Taps	0-1/5/10/25/50A (0-100% regulation per tap)
Duty	Continuous up to 10A; 25A & 50A: intermittent (5 min ON, 10 min OFF)
Test Voltage	0-110% of 110V ph-ph (63.5V ph-N), 3-phase, continuously variable
Fixed DC Voltage	Isolated 30V / 110V / 220V at 100 VA, available for 30 min
Phase Angle	0-360° - coarse 60° steps + fine ±30° (Unit 2, static phase-shifter)
Phase Angle Meter	360° range, 1° resolution
Time Measurement	0-9.999 sec (0.001s res.) and 0-99.99 sec (0.01s res.)
Current Metering	CT-operated ammeter per phase, 0-1A, 0-100% scaled, ±1% accuracy
Voltage Metering	0-200V, ±1% accuracy, 0.1V resolution, phase selector
Construction	Two-unit system; 1m interconnection cable; 4-core 5m mains cable
Warranty	12 months against manufacturing defects

02 Typical Applications

- Distance relay commissioning
- Directional and differential protection relay testing
- Over-current, earth fault, over/under-voltage relay testing
- Phase-to-phase fault simulation
- Indian Railways traction protection relay testing

03 What's in the Box

- ✓ Operation manual
- ✓ Schematic circuit diagram
- ✓ Works test report
- ✓ 1m interconnection cable (Unit 1 to Unit 2)
- ✓ 4-core, 5m mains input cable